

The Language of Medicine

13th edition

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Chapter 12

Respiratory System

Chapter Goals (Slide 1 of 2)

- Name the organs of the respiratory system and describe their location and function.
- Identify various pathologic conditions that affect the system.
- Learn medical terms that pertain to respiration.

Chapter Goals (Slide 2 of 2)

- Describe important clinical procedures related to the respiratory system and recognize relevant abbreviations.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 12

Lesson 12.1

Introduction (Slide 1 of 2)

External respiration: Exchange of air in lung capillaries

- Oxygen inhaled into the air sacs of the lungs
- Immediately passes into capillaries
- Carbon dioxide passes from capillaries into air sacs, to be exhaled

Introduction (Slide 2 of 2)

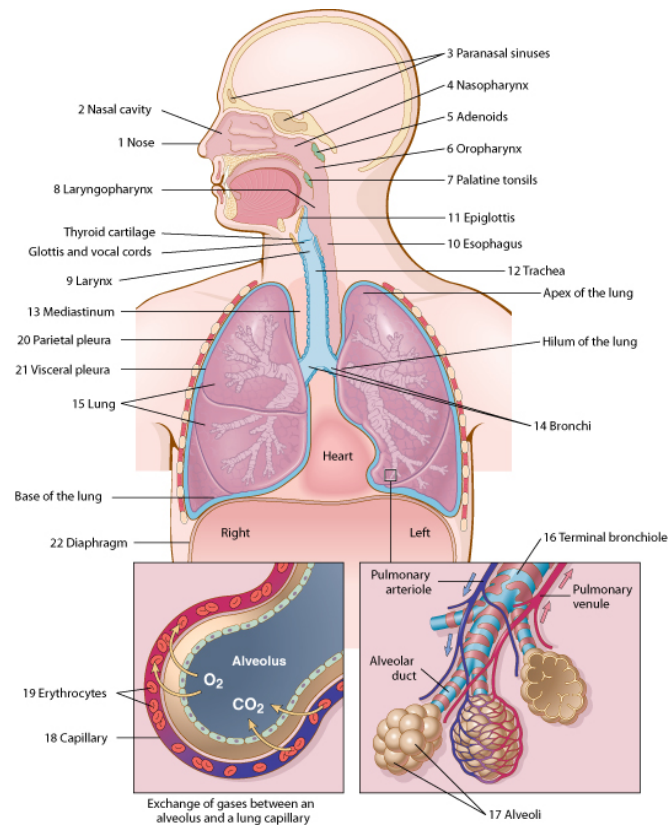
Internal respiration: Exchange of gases in the cells

- Occurs simultaneously between cells and capillaries
- Oxygen passes out of the bloodstream into the tissues
- Carbon dioxide passes out of tissues back into the bloodstream to travel to the lungs

Respiratory Anatomy

- **Mediastinum:** Middle of chest where trachea divides into two branches
- **Bronchi:** Each leads to a separate lung
- **Lungs:** Lobes are not mirror images
- **Hilum:** Blood vessels, nerves, lymphatic tissue, and bronchial tubes enter and exit

Anatomy of Respiration



Respiratory Structures

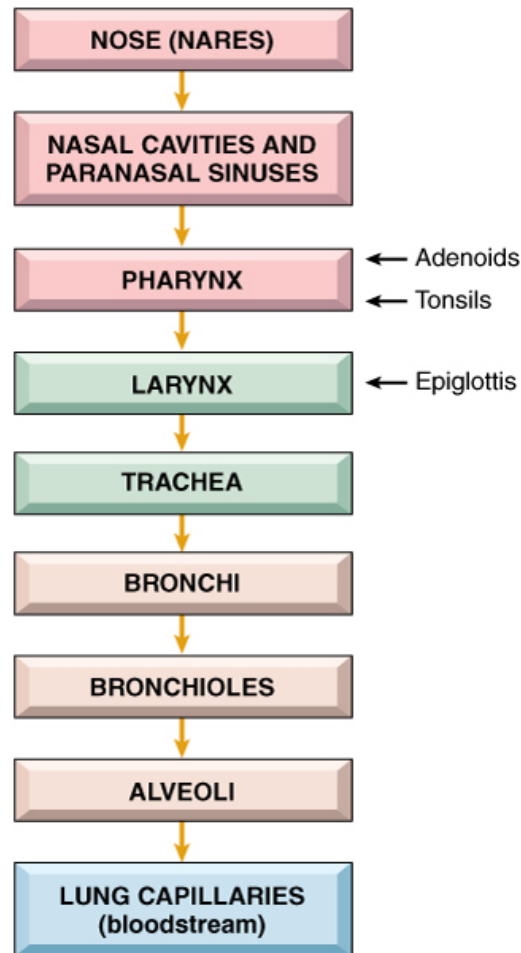
Conduct air

- Nose
- Nasal cavities and paranasal sinuses
- Pharynx
- Larynx
- Trachea

Exchange gases

- Bronchioles
- Alveoli
- Lung capillaries

Respiratory Flow Chart



Vocabulary (Slide 1 of 20)

Term
adenoids
alveolus (<i>plural</i> : alveoli)
apex of the lung

Vocabulary (Slide 2 of 20)

Term	Meaning/Definition
adenoids	Lymphatic tissue in the nasopharynx; pharyngeal tonsils
alveolus (<i>plural:</i> alveoli)	Air sac in the lung
apex of the lung	Tip or uppermost portion of the lung

Vocabulary (Slide 3 of 20)

Term
base of the lung
bronchioles
bronchus (<i>plural</i> : bronchi)

Vocabulary (Slide 4 of 20)

Term	Meaning/Definition
base of the lung	The lower portion of the lung
bronchioles	The smallest branches of the bronchi
bronchus (<i>plural</i> : bronchi)	The branch of the trachea (windpipe) that is a passageway into the lung; bronchial tube

Vocabulary (Slide 5 of 20)

Term
carbon dioxide (CO ₂)
cilia
diaphragm

Vocabulary (Slide 6 of 20)

Term	Meaning/Definition
carbon dioxide (CO ₂)	Gas produced by body cells when oxygen and carbon atoms from food combine; exhaled through the lungs
cilia	Thin hairs attached to mucous membrane epithelium lining the respiratory tract
diaphragm	Muscle separating the chest and abdomen; contracts to pull air into lungs; relaxes to push air out

Vocabulary (Slide 7 of 20)

Term
epiglottis
expiration
glottis

Vocabulary (Slide 8 of 20)

Term	Meaning/Definition
epiglottis	Lid-like piece of cartilage that covers the larynx; keeps food from entering the larynx and trachea during swallowing
expiration	Breathing out (exhalation)
glottis	Slit-like opening to the larynx

Vocabulary (Slide 9 of 20)

Term
hilum (of lung)
inspiration
larynx

Vocabulary (Slide 10 of 20)

Term	Meaning/Definition
hilum (of lung)	Midline region where the bronchi, blood vessels, and nerves enter and exit the lungs
inspiration	Breathing in (inhalation)
larynx	Voice box; contains the vocal cords

Vocabulary (Slide 11 of 20)

Term
lobe
mediastinum
nares

Vocabulary (Slide 12 of 20)

Term	Meaning/Definition
lobe	Division of a lung
mediastinum	The region between lungs in the chest cavity; contains the trachea, heart, lymph nodes, aorta, esophagus, and bronchial tubes
nares	Openings through the nose carrying air into the nasal cavities

Vocabulary (Slide 13 of 20)

Term
oxygen (O ₂)
palatine tonsil

Vocabulary (Slide 14 of 20)

Term	Meaning/Definition
oxygen (O ₂)	Gas that makes up 21% of air; passes into the bloodstream at the lungs and travels to all body cells
palatine tonsil	One of a pair of almond-shaped masses of lymphatic tissue in the oropharynx

Vocabulary (Slide 15 of 20)

Term
paranasal sinus
parietal pleura
pharynx

Vocabulary (Slide 16 of 20)

Term	Meaning/Definition
paranasal sinus	One of the air cavities in the bones near the nose
parietal pleura	Outer fold of pleura lying closer to the ribs and chest wall
pharynx	The throat; including the nasopharynx, oropharynx, and laryngopharynx

Vocabulary (Slide 17 of 20)

Term
pleura
pleural cavity
pulmonary parenchyma

Vocabulary (Slide 18 of 20)

Term	Meaning/Definition
pleura	Double-folded membrane surrounding each lung
pleural cavity	Space between the folds of the pleura
pulmonary parenchyma	Essential parts of the lung, responsible for respiration; bronchioles and alveoli

Vocabulary (Slide 19 of 20)

Term
respiration
trachea
visceral pleura

Vocabulary (Slide 20 of 20)

Term	Meaning/Definition
respiration	The process of moving air into and out of the lungs; breathing
trachea	Windpipe
visceral pleura	Inner fold of pleura lying closer to the lung tissue

QUICK QUIZ

1. What is the tip or uppermost portion of the lung?

- A. Base
- B. Lobe
- C. Apex
- D. Nares

Combining Forms and Terminology

(Slide 1 of 5)

Combining Form	Meaning
adenoid/o	adenoids
alveol/o	alveolus, air sac
bronch/o	bronchial tube
bronchi/o	bronchus
bronchiol/o	bronchiole, small bronchus
capn/o	carbon dioxide

Combining Forms and Terminology

(Slide 2 of 5)

Combining Form	Meaning
coni/o	dust
cyan/o	blue
epiglott/o	epiglottis
laryng/o	larynx, voice box
lob/o	lobe of the lung
mediastin/o	mediastinum

Combining Forms and Terminology

(Slide 3 of 5)

Combining Form	Meaning
nas/o	nose
orth/o	straight, upright
ox/o	oxygen
pector/o	chest
pharyng/o	pharynx, throat
phon/o	voice

Combining Forms and Terminology

(Slide 4 of 5)

Combining Form	Meaning
phren/o	diaphragm
pleur/o	pleura
pneum/o	air, lung
pneumon/o	lung
pulmon/o	lung
rhin/o	nose

Combining Forms and Terminology

(Slide 5 of 5)

Combining Form	Meaning
sinus/o	sinus cavity
spir/o	breathing
tel/o	complete
thorac/o	chest
tonsill/o	tonsils
trache/o	trachea, windpipe

Combining Forms, Suffixes, and Terminology (Slide 1 of 1)

Suffix	Meaning
-ema	condition
-osmia	smell
-pnea	breathing
-ptysis	spitting
-sphyxia	pulse
-thorax	pleural cavity, chest

Chapter 12

Lesson 12.2

Diagnostic Terms

- Auscultation
- Percussion
- Pleural rub
- Rales (crackles
- Rhonchi (singular rhonchus)
- Sputum
- Stridor
- Wheezes

Pathology (Slide 1 of 4)

Upper respiratory disorders

- Croup
- Diphtheria
- Epistaxis
- Pertussis

Pathology (Slide 2 of 4)

Bronchial tube disorders

- Asthma
- Bronchiectasis
- Chronic bronchitis
- Cystic fibrosis

Pathology (Slide 3 of 4)

Lung disorders

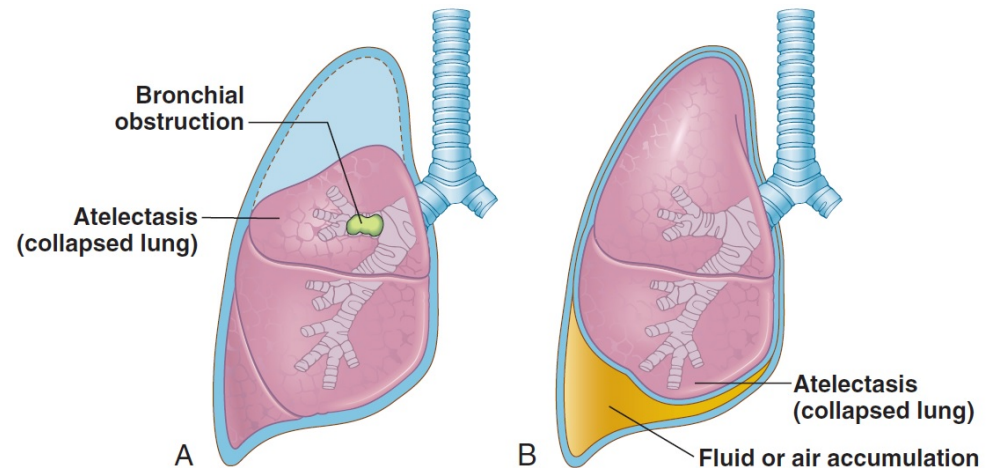
- Atelectasis
- Emphysema
- Lung cancer
- Pneumoconiosis
- Pneumonia
- Pulmonary abscess
- Pulmonary edema
- Pulmonary embolism (PE)
- Pulmonary fibrosis
- Sarcoidosis
- Tuberculosis (TB)

Pathology

Lung Disorders

Atelectasis

- Two forms
- Proximal obstruction of distal airways
- Accumulations of fluid, blood, or air in the pleural cavity



Pathology (Slide 4 of 4)

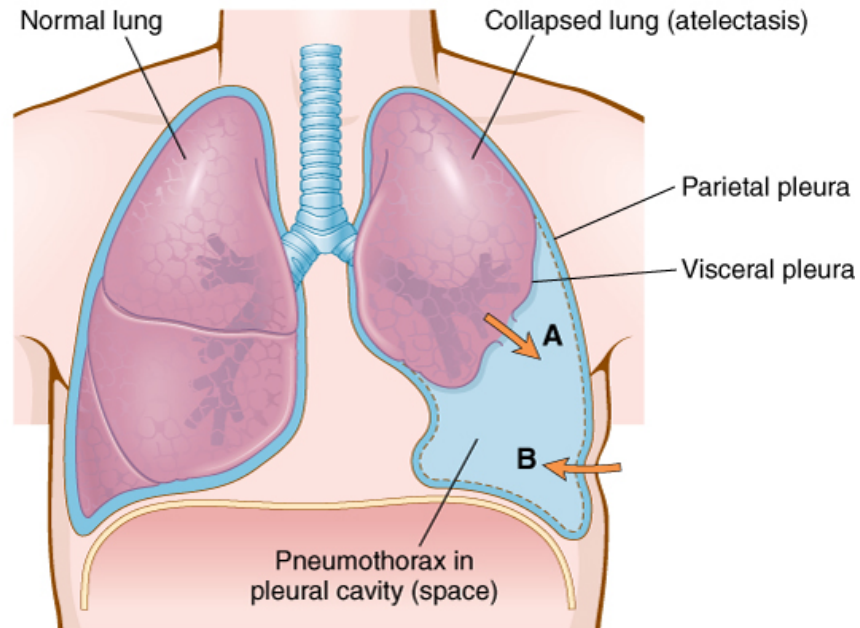
Pleural disorders

- Mesothelioma
- Pleural effusion
- Pleurisy (pleuritis)
- Pneumothorax

Pathology

Pleural Disorders

Pneumothorax: Air gathers in the pleural cavity



Study Section

- Anthracosis
- Asbestosis
- Bacilli
- Chronic obstructive pulmonary disease (COPD)
- Cor pulmonale
- Exudate
- Hydrothorax
- Infiltrate
- Palliative
- Paroxysmal
- Pulmonary infarction
- Purulent
- Silicosis

Chapter 12

Lesson 12.3

Clinical Procedures (Slide 1 of 9)

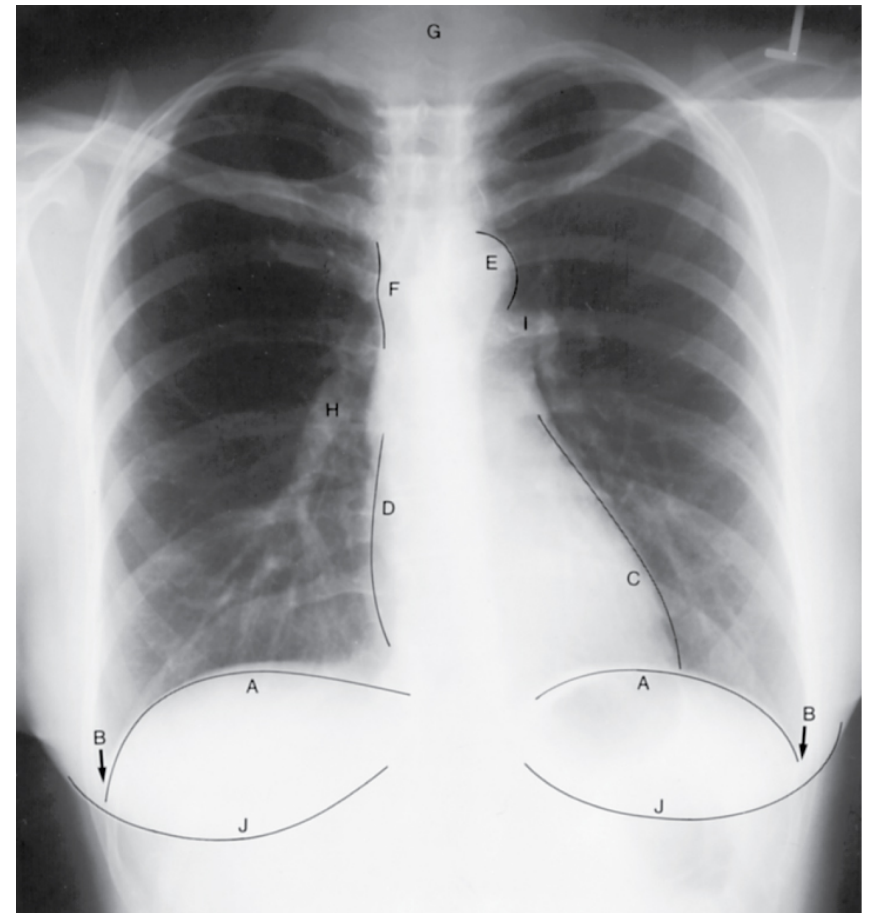
X-rays

- Chest x-ray (CXR)
- Computed tomography (CT) scan of the chest
 - Pulmonary angiography

Clinical Procedures (Slide 2 of 9)

X-rays: radiographic imaging

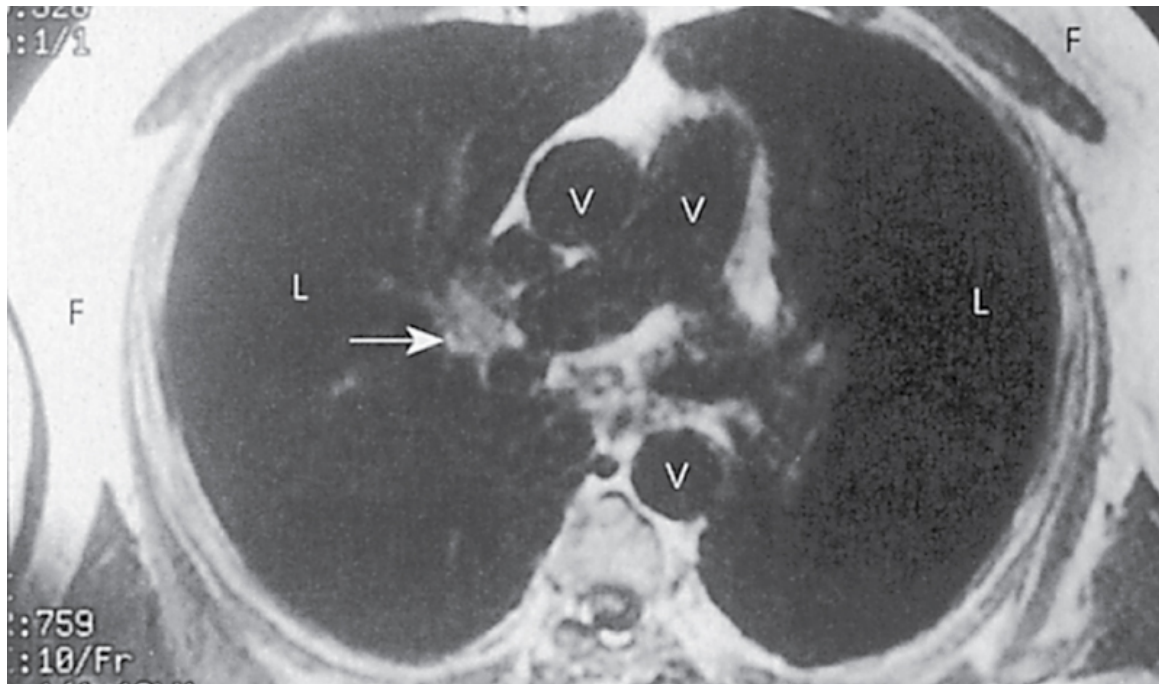
This is a normal chest x-ray taken from the PA (posteroanterior) view—x-ray passes from back to front.



Clinical Procedures (Slide 3 of 9)

Magnetic resonance imaging (MRI):

- magnetic waves create images

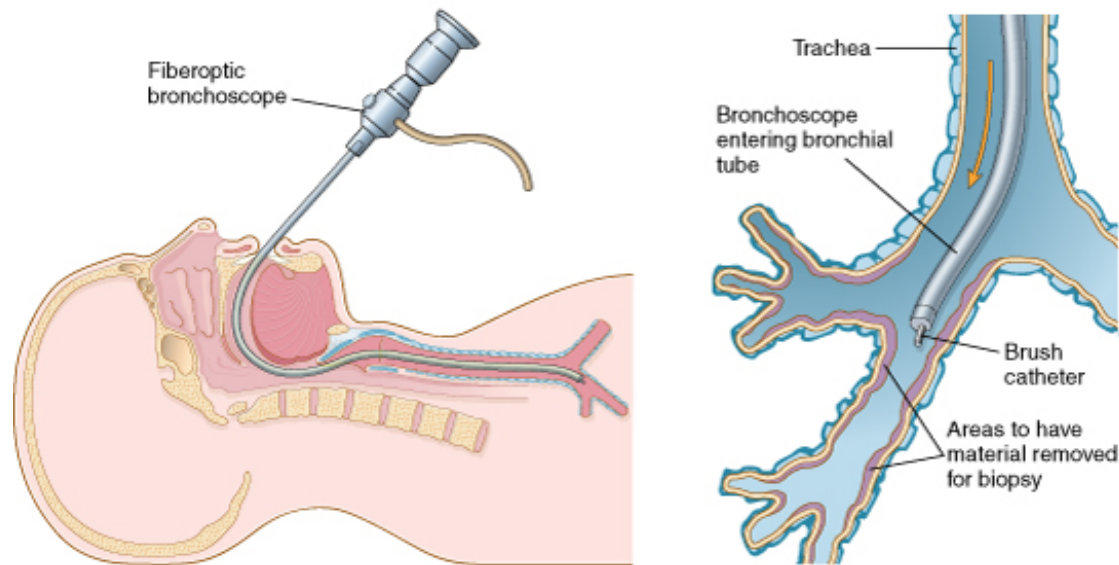


Clinical Procedures (Slide 4 of 9)

- **Nuclear tests**
 - **Positron emission tomography (PET) scan of the lung**
 - **Ventilation-perfusion (V/Q) scan:** inhalation of radioactive gas for imaging, followed by injection of different radioactive material for imaging

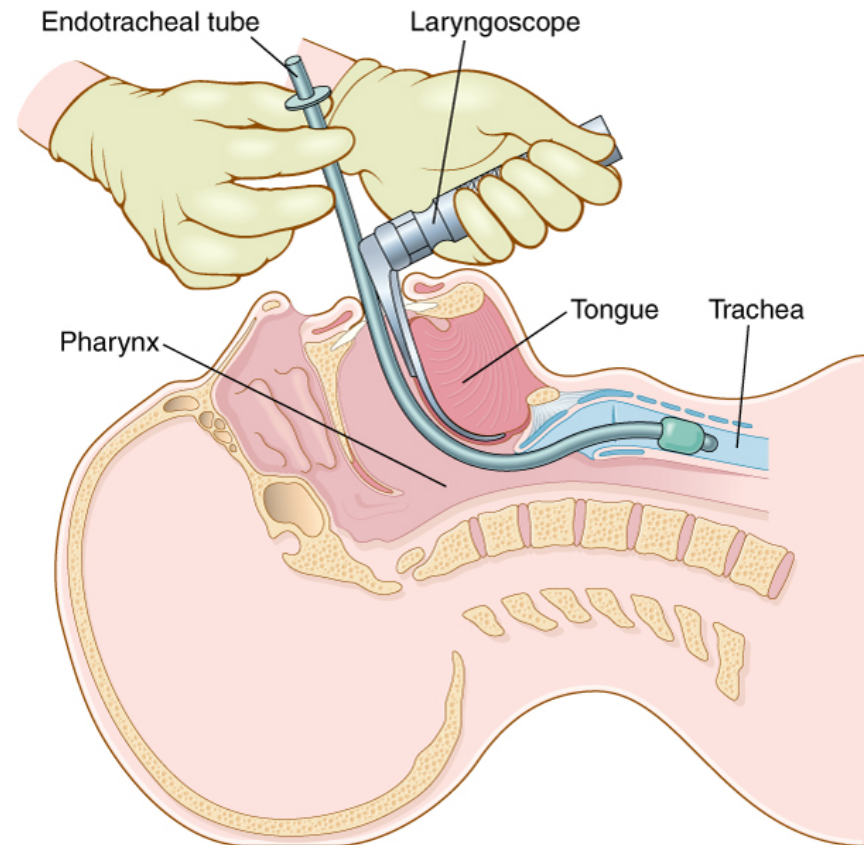
Clinical Procedures (Slide 5 of 9)

Bronchoscopy: fiberoptic or rigid endoscope inserted into the bronchial tubes for diagnosis, biopsy, or collection of secretions



Clinical Procedures (Slide 6 of 9)

Endotracheal intubation:
placement of tube from
mouth to trachea to
establish airway

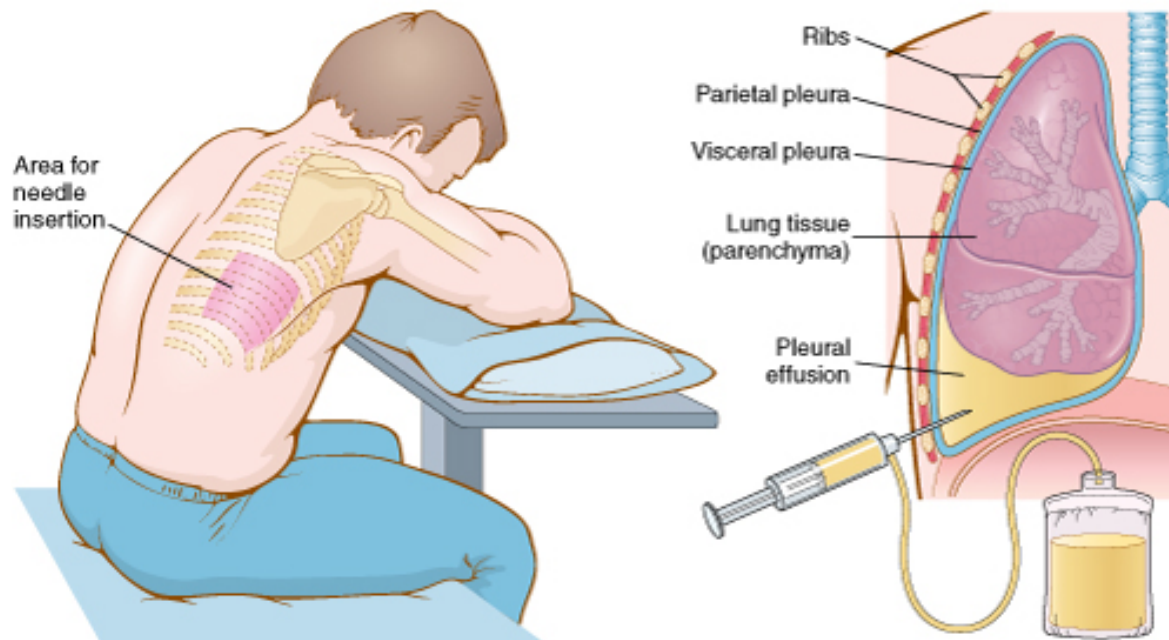


Clinical Procedures (Slide 7 of 9)

- Laryngoscopy
- Lung biopsy
- Mediastinoscopy
- Pulmonary function tests (PFTs)
- Thoracentesis
- Thoracotomy
- Thorascopy
- Tracheostomy
- Tuberculin test
- Tube thoracostomy

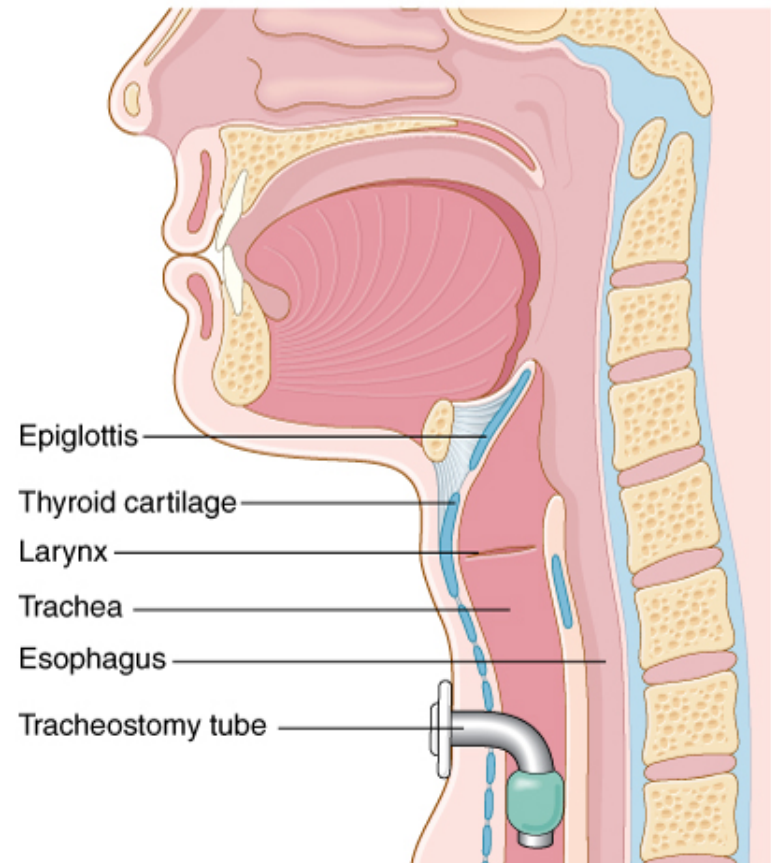
Clinical Procedures (Slide 8 of 9)

Thoracentesis: surgical puncture to remove fluid from pleural space



Clinical Procedures (Slide 9 of 9)

Tracheostomy: surgical creation of an opening into the trachea through the neck



Abbreviations (Slide 1 of 5)

Abbreviation	Meaning
ABGs	Arterial blood gases
AFB	Acid-fast bacillus
ARDS	Acute respiratory distress syndrome
BAL	Bronchioalveolar lavage
Bronch	Bronchoscopy
CF	Cystic fibrosis
CO ₂	Carbon dioxide
COPD	Chronic obstructive pulmonary disease
CPAP	Continuous positive airway pressure
CPR	Cardiopulmonary resuscitation

Abbreviations (Slide 2 of 5)

Abbreviation	Meaning
C & S	Culture and sensitivity testing (of sputum)
CTPA	Computed tomography pulmonary angiography
CXR	Chest x-ray (film)
DI _{CO}	Diffusion capacity of lung for carbon monoxide
DOE	Dyspnea on exertion
DPT	Diphtheria, pertussis, tetanus
FEV ₁	Forced expiratory volume in 1 second
FVC	Forced vital capacity
ICU	Intensive care unit

Abbreviations (Slide 3 of 5)

Abbreviation	Meaning
LLL	Left lower lobe (of lung)
LUL	Left upper lobe (of lung)
MAC	Mycobacterium avium
MDI	Metered-dose inhaler
NSCLC	Non-small cell lung cancer
O ₂	Oxygen
OSA	Obstructive sleep apnea
Pa ^{CO2}	Carbon dioxide partial pressure
Pa ^{O2}	Oxygen partial pressure
PCP	Pneumocystis pneumonia
PE	Pulmonary embolism

Abbreviations (Slide 4 of 5)

Abbreviation	Meaning
PEP	Positive expiratory pressure
PEEP	Positive end-expiratory pressure
PFTs	Pulmonary function tests
PND	Paroxysmal nocturnal dyspnea
PPD	Purified protein derivative
RDS	Respiratory distress syndrome
RLL	Right lower lobe (of lung)
RML	Right middle lobe
RSV	Respiratory syncytial virus
RUL	Right upper lobe (of lung)
RV	Residual volume

Abbreviations (Slide 5 of 5)

Abbreviation	Meaning
SABA	Short-acting beta agonist
SCLC	Small cell lung cancer
SOB	Shortness of breath
TB	Tuberculosis
TLC	Total lung capacity
URI	Upper respiratory infection
V _T	Tidal volume
VATS	Video-assisted thoracic surgery
VC	Vital capacity
V/Q scan	Ventilation-perfusion scan

Review Sheet – Combining Forms (Slide 1 of 10)

Combining Form
adenoid/o
alveol/o
bronch/o
bronchi/o
bronchiol/o
capn/o

Review Sheet – Combining Forms (Slide 2 of 10)

Combining Form	Meaning
adenoid/o	adenoids
alveol/o	alveolus; air sac; small sac
bronch/o	bronchial tube
bronchi/o	bronchial tube
bronchiol/o	bronchiole
capn/o	carbon dioxide

Review Sheet – Combining Forms (Slide 3 of 10)

Combining Form
coni/o
cyan/o
epiglott/o
laryng/o
lob/o
mediastin/o

Review Sheet – Combining Forms (Slide 4 of 10)

Combining Form	Meaning
coni/o	dust
cyan/o	blue
epiglott/o	epiglottis
laryng/o	larynx (voice box)
lob/o	lobe
mediastin/o	mediastinum

Review Sheet – Combining Forms (Slide 5 of 10)

Combining Form
nas/o
orth/o
ox/o
pector/o
pharyng/o
phon/o

Review Sheet – Combining Forms (Slide 6 of 10)

Combining Form	Meaning
nas/o	nose
orth/o	straight
ox/o	oxygen
pector/o	chest
pharyng/o	throat (pharynx)
phon/o	voice; sound

Review Sheet – Combining Forms (Slide 7 of 10)

Combining Form
phren/o
pleur/o
pneum/o
pneumon/o
pulmon/o
rhin/o

Review Sheet – Combining Forms (Slide 8 of 10)

Combining Form	Meaning
phren/o	diaphragm; mind
pleur/o	pleura
pneum/o	lung; air; gas
pneumon/o	lung; air; gas
pulmon/o	lung
rhin/o	nose

Review Sheet – Combining Forms (Slide 9 of 10)

Combining Form
sinus/o
spir/o
tel/o
thorac/o
tonsill/o
trache/o

Review Sheet – Combining Forms (Slide 10 of 10)

Combining Form	Meaning
sinus/o	sinus
spir/o	to breathe
tel/o	complete
thorac/o	chest
tonsill/o	tonsil
trache/o	trachea (windpipe)

Review Sheet – Suffixes (Slide 1 of 2)

Suffix
-ema
-osmia
-pnea
-ptysis
-sphyxia
-thorax

Review Sheet – Suffixes (Slide 2 of 2)

Suffix	Meaning
-ema	condition
-osmia	smell
-pnea	breathing
-ptysis	spitting
-sphyxia	pulse
-thorax	chest; pleural cavity